When Auditing the Unobserved Objective Doesn't Help: A Calibrated Negative Result in Proxy-Optimizing Incentive Allocation

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Abstract

A learning system that allocates employee incentives is naturally rewarded on what it can log—retention—not on the welfare it cannot see. We build an open, CPU-only testbed for this and confirm the expected harm: a retention-optimizer erodes latent satisfaction and, more sharply, cross-archetype *equity*, robustly across 250 randomized worlds and under calibration to the IBM HR Analytics benchmark. The natural remedy is to *audit* the hidden objective (survey a fraction p, fold it into the reward). We were excited to find that sparse auditing *backfires*—worse than no audit. This note reports, in full, why that exciting result did not survive: two pre-registered kill-tests show the backfire is an artifact of a naive unshrunk estimator and of an implausibly strong satisfaction–retention coupling, vanishing under a precision-aware estimator *or* the weaker coupling the benchmark actually exhibits. The durable, un-flashy lesson is distributional—measure the true objective's *distribution*, not its average—and the audit remedy is a near-null we report rather than bury.

1 Setup

We simulate N=400 employees over T=12 periods, four equal-or-calibrated archetypes, and five incentive categories under a budget. Each employee has a latent satisfaction $s \in [0, 1]$ (unitless, illustrative). Attrition follows $\Pr(\text{leave}) = \sigma(\beta_0 - \beta_s s - h\mathbb{1}[\text{recent monetary}])$, where h is a "handcuff" that raises retention *without* raising satisfaction (vesting/options). Six allocators are compared: Random, Monetary-only, a theory-based **Static** baseline, a retention-rewarded **Learner** (ε-greedy bandit), a **Learner+audit** that folds a survey of fraction p into its reward through a satisfaction estimator $\phi \in \{\texttt{raw}, \texttt{shrink}, \texttt{gated}\}$, and an **Oracle**. We report paired-by-seed differences (200 seeds) with 95% CIs and per-seed sign-stability; we do not headline standardized effect sizes (they measure within-model separation in a seed-controlled sim). Code, data, kill-tests, and an AI-use log are released.

2 The robust finding (background)

The Learner matches Static on *observed* retention (0.941 vs. 0.941 at h=1) while its *latent* satisfaction is far lower (0.529 vs. 0.566; -0.037 [± 0.002], 100% sign-stable) and its cross-archetype inequality is $\sim 4\times$ larger. This persists at zero crowding-out ($+0.015$), on a never-left founder cohort ($+0.014$; ruling out survivorship composition), in 100% of 250 randomized worlds, and under calibration to IBM HR Analytics (n=1470): the fitted hazard slope is 0.75–0.95 (vs. our illustrative 3.2), the archetype mix is skewed (low-comp 39%), and an off-satisfaction retention channel is empirically present (stock-option coefficient -0.51 [$-0.72, -0.31$] controlling for satisfaction). We read this as a

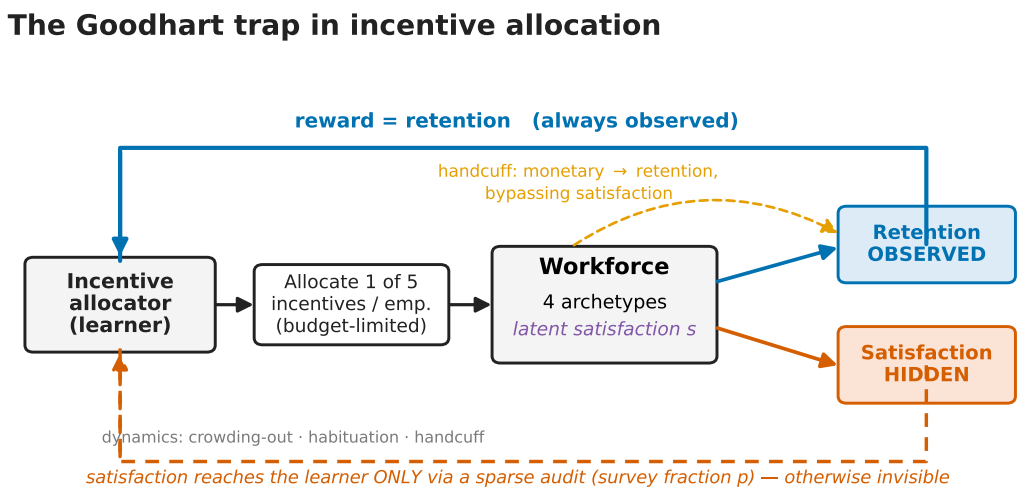


Figure 1: *(AI-generated.)* The reward sees only retention; satisfaction returns only through a sparse audit of fraction p. The handcuff channel decouples the optimized proxy from the true objective.

distributional sharpening of multitask agency [1]—known in the aggregate, but the equity axis and the calibration are the contribution.

3 The negative result

The natural fix is to audit satisfaction and reward the learner on it. With a naive `raw` estimator we found a sharp backfire: auditing at p=0.05 gave *lower* satisfaction than no audit (-0.0085, $p_{\text{FDR}}<10^{-13}$ at h=0), as a low-precision estimate ($\sim$1 survey per cell per period) enters the argmax at full weight. Two pre-registered falsification tests dismantle the strong claim (Fig. 2):

- **Estimator (K3).** A precision-weighted `shrink` estimator removes the backfire at both h ($+0.0003$, $+0.0006$; n.s. after FDR).
- **Calibration.** Under the fitted (weaker) hazard slope the backfire vanishes even for `raw` ($-0.008 \to +0.003$): it required a strong coupling the benchmark does not exhibit.

Across 250 randomized worlds the raw backfire holds in only 62% (vs. 100% for the welfare/equity gap). The honest conclusion: naive auditing backfires only in a narrow regime (strong coupling $\times$ naive estimator); otherwise auditing the hidden objective is roughly neutral. The remedy is not the story.

4 Takeaways

(1) A proxy-optimizing allocator erodes unobserved welfare and, more sharply, its *distribution* across worker types—robustly and under real-data calibration; track per-cohort welfare, because the aggregate hides where harm lands. (2) The intuitive audit remedy is, honestly, a near-null under realistic conditions; the exciting backfire we first saw was a fragile artifact, and we report it as

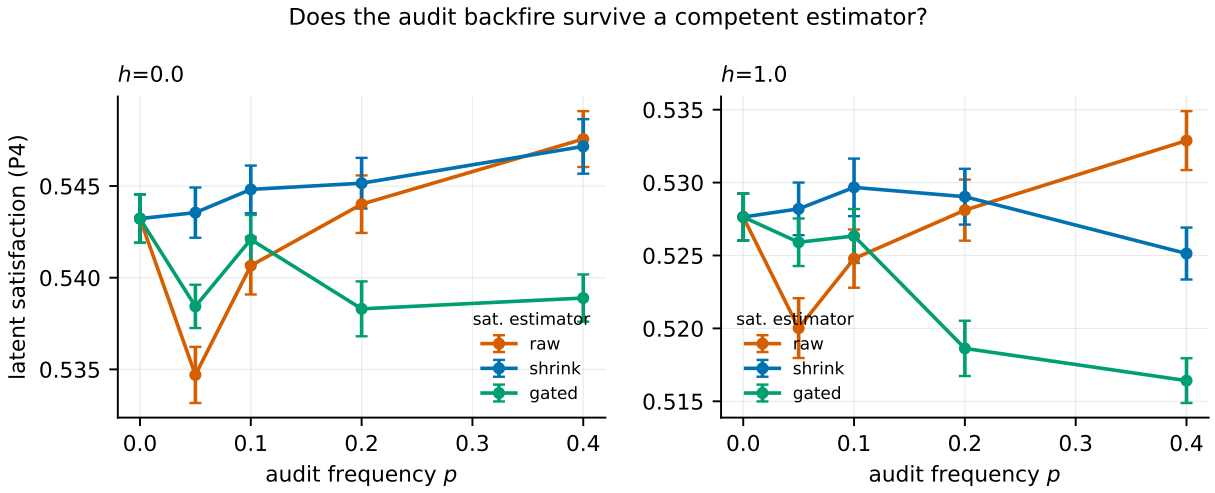


Figure 2: *(AI-generated.)* The audit backfire is fragile. Audited-learner satisfaction vs. audit frequency p under three estimators; the low-p dip under `raw` is removed by `shrink` (n.s. after FDR), and also vanishes under benchmark-calibrated coupling (not shown; $-0.008 \to +0.003$). 200 seeds, 95% CI.

such. Following the negative-results ethos [2], we hope the failed remedy is as useful to others as a positive one would have been. *Limitations:* the dynamics (gains, crowding, habituation) remain illustrative; the benchmark is synthetic and cross-sectional; the learner is a cohort bandit. Full paper, all kill-tests, and calibration are in the repository.

References

[1] Holmström, B., & Milgrom, P. (1991). Multitask Principal-Agent Analyses. *J. Law, Econ., & Org.*, 7(SI), 24–52.

[2] Karwowski, J. et al. (2024). Goodhart's Law in Reinforcement Learning. *ICLR*. (See also: Embracing Negative Results in ML, arXiv:2406.03980.)